

Beyond Representation: Index Structure Limitations in Dense Retrieval

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Why this talk

- ▶ Dense retrieval is everywhere: e-commerce, enterprise search, RAG.
- ▶ Practical deployments rely on **approximate nearest-neighbor (ANN)** indices (HNSW, IVF) for speed.
- ▶ Most research focuses on the *representation*—can the embedding capture the right semantics?
- ▶ This talk: **the index itself imposes a separate, quantifiable failure mode** that disproportionately attacks a specific population of queries.

Headline finding

Even when the embedding has enough representational capacity, switching from exact (Flat) to HNSW search forfeits $\sim 63\%$ **of achievable Recall@10** on identifier queries—vs. less than 2% on standard queries.

The two camps

Sparse retrieval (BM25)

- ▶ Inverted index over tokens.
- ▶ Exact lexical matching.
- ▶ Strong on rare terms, IDs, named entities.

Dense retrieval

- ▶ Continuous embedding + nearest-neighbor search.
- ▶ Strong semantic generalization.
- ▶ Indispensable in production for fluency & multilinguality.

- ▶ Prior theory (Tay et al. 2020; Fan et al. 2022): *representation capacity* of dense models can match sparse—given enough dimensions.
- ▶ But: practical dense retrieval uses ANN, not exact search.

What does that cost?

Definition (operational)

Queries whose relevance hinges on a single, low-frequency, high-IDF token: identifiers (SKUs, part numbers), rare terms, precise named entities.

- ▶ In production: a small fraction of total traffic.
- ▶ But operationally critical: failures yield empty search-result pages, abandoned carts, broken RAG retrievals.
- ▶ In our experiments:
 - ▶ **ID subset**: doc identifier used as query (extreme case).
 - ▶ **Rare subset**: uni/bigram terms appearing in ≤ 5 docs.
 - ▶ **Standard subset**: real user queries (control).

The mechanism: centroid dilution (theory)

IVF clusters the corpus into N cells; at query time only the closest n_{probe} centroids are searched.

$$\text{Centroid of cluster } C: \quad c = \frac{1}{|C|} \sum_{d \in C} f(d) = \dots + \alpha_{t^*} p \phi(t^*) + \dots$$

For a query dominated by rare token t^* :

$$\langle c, q \rangle \approx \alpha_{t^*}^2 p = \log^2\left(\frac{M}{n_{t^*}}\right) \cdot \frac{n_{t^*, C}}{M/N}$$

- ▶ α_{t^*} is large for rare tokens (high IDF).
- ▶ But p (in-cluster prevalence) is small for rare tokens.
- ▶ JL projection adds noise floor $\epsilon \approx \sqrt{\log V/d}$.
- ▶ Rare-token centroid signal lost when $\alpha_{t^*}^2 p \lesssim \epsilon$.

Larger embedding d helps ϵ , but cannot fix small p .

Experimental setup

Datasets

- ▶ **Home Depot:** 54,667 product titles
- ▶ **MS MARCO Passage v1.1:** 82,360 passages

Three query subsets per dataset

(1,000 queries each)

- ▶ Standard | Rare (≤ 5 docs) | ID

Representation

- ▶ Char-trigram + JL projection
- ▶ $d \in \{256, 512, 1024\}$
- ▶ Unsupervised

Indices (FAISS)

- ▶ Flat (exact)
- ▶ IVF (nlist, nprobe sweep)
- ▶ HNSW ($M=16$, efC=200, efS sweep)
- ▶ BM25 (Pyserini)

Why unsupervised trigram, not DPR / Sentence-BERT?

A supervised encoder abstracts away the lexical detail that *defines* a spear-fishing query. The Flat baseline would already fail on IDs—conflating representation failure with indexing failure. We need a representation that *provably* preserves lexical distances (Johnson–Lindenstrauss).

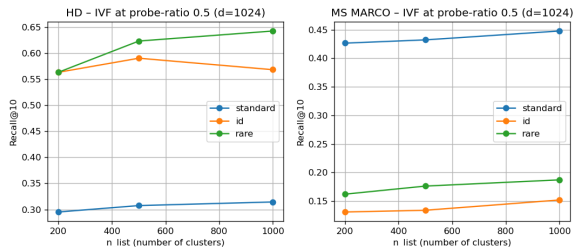
Headline result: the gap

Dataset	Subset	Flat	IVF	HNSW	BM25
HD	standard	0.327	0.314	0.321	0.447
HD	rare	0.751	0.642	0.619	0.878
HD	id	0.874	0.590	0.327	1.000
MS MARCO	standard	0.471	0.448	0.446	0.770
MS MARCO	rare	0.205	0.187	0.166	0.688
MS MARCO	id	0.174	0.152	0.071	—

Recall@10 at $d=1024$, across-token trigrams, best ANN sweep config.

- ▶ Flat→HNSW gap (relative): HD {standard **1.8%**, rare **17.5%**, **id 62.6%**}; MM {standard **5.3%**, rare **19.0%**, **id 59.2%**}.
- ▶ Paired-bootstrap 95% CI on Flat—HNSW excludes zero everywhere (HD-ID +0.576, MM-ID +0.138).
- ▶ *MS MARCO caveat*: BM25 > Flat dense on standard/rare reflects a representation limit of unsupervised trigrams—our claim is the Flat→ANN *gap*, well-defined at any absolute level.

Indirect evidence: IVF cluster granularity



Theory predicts: smaller clusters $\Rightarrow$ less centroid dilution $\Rightarrow$ rare/ID queries benefit more from finer clustering.

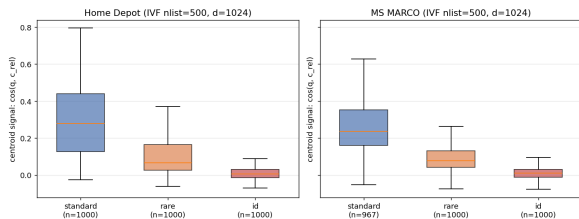
Data: rare & ID gain $2-3\times$ *more* (relative) than standard going from $n_{\text{list}}=200 \rightarrow 1000$.

Asymmetric improvement is exactly what the centroid-dilution bound (Eq. 1) predicts.

Direct evidence: measure the centroid signal

For each query, compute the centroid signal $\langle q, c_{\text{rel}} \rangle$ on the centroid of the query's relevant cluster (IVF, $n_{\text{list}}=500$, $d=1024$).

Centroid signal for relevant cluster, by query subset



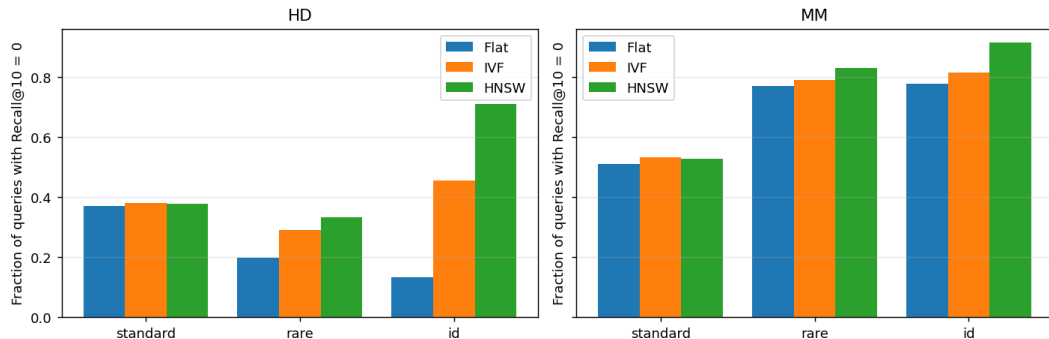
Subset (HD)	signal	rank	hit@250
standard	0.279	2	97%
rare	0.066	69	81%
id	0.007	216	59%

~ 40 $\times$ collapse from standard to ID:
centroid signal disappears into the JL noise floor exactly as the bound (Eq. 1) predicts.

For HD-ID queries, 41% of relevant clusters fall outside the top-half by centroid score—*structurally unrecoverable*.

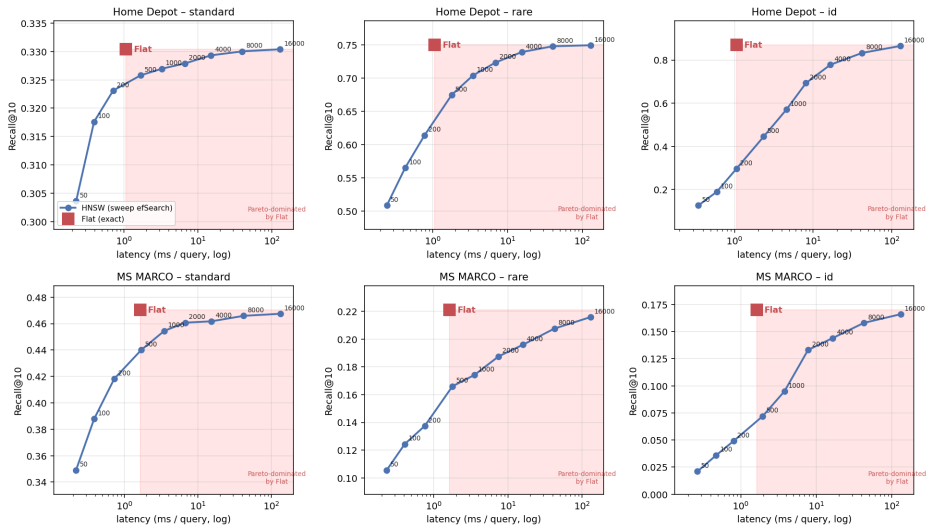
The empty-result rate

Empty-result rate: fraction of queries that retrieve ZERO relevant docs in top-10



- ▶ HD-id Flat **13%** → HNSW **71%** return nothing relevant in top-10.
- ▶ MM-id Flat **78%** → HNSW **92%**.
- ▶ Standard subsets: ≤ 2 percentage points either way on both datasets.
- ▶ *This is the production-visible symptom.*

The recall-latency trade-off changes shape



HNSW efSearch sweep (standard FAISS recipe $M=16$, $efC=200$) vs. Flat (red square). Red region = strictly Pareto-dominated by Flat.

The cost-to-close ratio (corpus-invariant)

efSearch needed to reach 95% of Flat's Recall@10:

Subset	efS	ms/q	vs. Flat
HD-standard	100	0.39	2.7× faster
HD-rare	2,000	6.94	6.5× slower
HD-id	8,000	41.9	40× slower
MM-standard	1,000	3.48	2.1× slower
MM-rare	16,000	129.6	79× slower
MM-id	16,000	132.0	81× slower

- ▶ Standard queries: HNSW gives the canonical ANN speedup.
- ▶ ID queries: need 80–160× more efSearch budget than standard.
- ▶ Search cost scales $\sim$ linearly in efSearch $\Rightarrow$ this ratio *generalizes to any corpus size*.

“But this is only because the embeddings are simple”

Reviewer question we anticipated: *do learned dense encoders avoid this?*

Re-ran the full grid with all-MiniLM-L6-v2 (Sentence-BERT, 384-dim, supervised).

Dataset	Subset	Trigram Flat R@10	MiniLM Flat R@10
HD	standard	0.327	0.298
HD	rare	0.751	0.428
HD	id	0.874	0.010
MS MARCO	standard	0.471	0.894
MS MARCO	rare	0.205	0.226
MS MARCO	id	0.174	0.001

- ▶ MiniLM nearly doubles MS MARCO standard recall—as designed.
- ▶ MiniLM **cannot distinguish identifiers**: “12345” $\approx$ “67890”.
- ▶ Representation collapses *before* the index has a chance to bias.
- ▶ This is exactly why a supervised encoder is the *wrong instrument* for our study—and exactly why hybrid retrieval is needed in production.

What this means for production

Two ways to read the gap

- ▶ **Default efSearch:** HNSW forfeits $\sim 63\%$ of achievable recall on identifier queries, $\sim 18\%$ on rare, $\sim 2\%$ on standard. *Visible:* 71% of HD-ID HNSW queries return nothing relevant.
- ▶ **Match Flat recall:** on ID you must pay 40–80 $\times$ Flat latency. ANN's speedup advantage *evaporates* on the population that needs reliability the most.

Recommendations:

1. *Query-type-aware routing:* send spear-fishing traffic to BM25 / Flat.
2. *Index-aware training:* penalize placement of rare features in poorly-reachable embedding regions.
3. *Hybrid retrieval:* sparse for precision on rare tokens, dense for semantic generalization—now with a quantitative justification.

Takeaways

1. Representation capacity is *necessary but not sufficient* for production dense retrieval.
2. ANN indices introduce a *second*, structurally different failure mode concentrated on rare-token / identifier queries.
3. We formalized this as **indexing bias** + **centroid-dilution bound**, then directly measured the centroid signal per query ($\sim 40\times$ collapse from standard to ID).
4. Reaching Flat-equivalent recall on ID requires $80\text{--}160\times$ more efSearch budget than on standard—the corpus-invariant quantification of the ANN value-proposition collapse.
5. Supervised encoders don't escape this—they collapse at the representation stage on the same queries.

Code & experiments: github.com/alemagnani/sparse-dense-paper

Thank you

Questions?

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